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Finite element modelling and design of concrete wind turbine towers subjected to combined compression and bending

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Abstract: The combined compression and bending condition is primarily considered in the design of concrete wind turbine towers. However, the design methods specified in existing standards fail to accurately predict the compression-bending capacity of the concrete tower upon the occurrence of cracks at horizontal joints. Therefore, this study aims to develop a new method for calculating the compression-bending capacity of concrete wind turbine towers. A finite element (FE) model was firstly developed and validated against experimental results regarding to the load-displacement curves, compression-bending capacities and failure modes. Upon validation, a parametric study was conducted to generate 125 numerical results and investigate the effects of key parameters, including concrete strength, yielding strength of the steel reinforcement, precompression ratio, cross-sectional dimensions, and prestressing force. The results indicated that the compression-bending capacity of concrete wind turbine towers are significantly influenced by the concrete strength, cross-sectional dimensions, and precompression ratio. Based on the newly generated results, new design rules are proposed and compared against both experimental and numerical results. Subsequently, the applicability of the existing design methods of compression-bending capacity for concrete wind turbine towers was evaluated; unfortunately, they were found to be unsafe due to the neglect of the increment of prestressing force and the change of the cross-sectional second moment of inertia. Finally, the new design method yields a mean ratio of predicted to experimental (and simulated) compression-bending capacity of 1.01, with a coefficient of variation (CoV) of 0.0454, demonstrating enhanced accuracy and consistency.

Keyword: Compression-bending capacity, Finite element, Horizontal joint, Prestressed concrete

tower, Wind turbine.

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1 Introduction

Wind energy is a clean and renewable energy source offering significant potential to mitigate the current energy shortfall and to achieve net-zero carbon emissions. Currently, the heights of wind turbine towers in onshore low-wind-speed areas typically exceed 140 meters [1-3]. Due to the high overall stiffness and low construction and maintenance costs, the hybrid tower structure is becoming the most popular type for onshore wind turbines [4-6]. Concrete wind turbine towers are typically constructed using prefabricated concrete ring segments through a modular assembly method. The concrete shells are connected by arc-shaped bolts and epoxy resin to form the ring segment. Afterwards, the ring segment are connected by epoxy resin, with the joints between ring segments are termed as horizontal joints [7-9]. Therefore, the concrete wind turbine towers contain a great number of connection joints and the reinforcement is discontinuous at these joints. The structure of steel-concrete hybrid tower is shown in Fig 1.

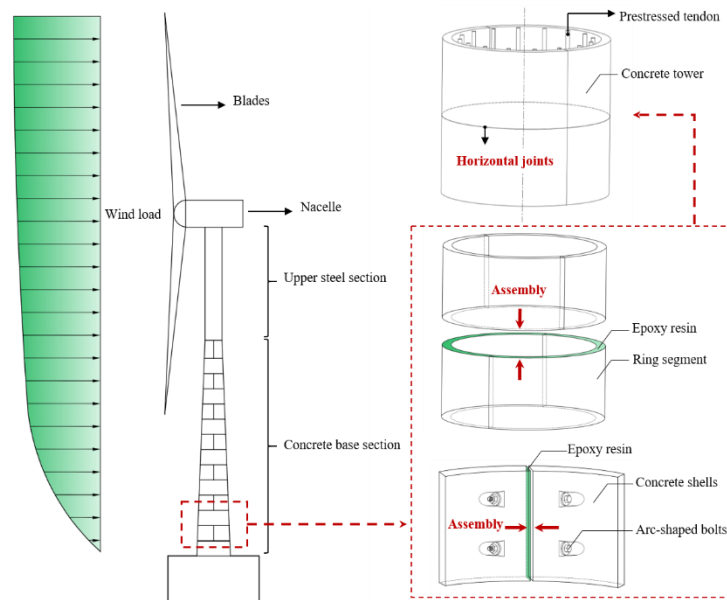


Fig 1 Schematic drawing of the steel-concrete hybrid tower.

Under wind load, the prestressed concrete towers are subjected to a combination of compression, bending, shear and torsion. Among them, the compression-bending load is the

governing one [10-11]. When ultimate bending moment causes horizontal joint cracking, the compression-bending capacity of the tower is primarily provided by the concrete in the compression zone because the reinforcement at the horizontal joints is discontinuous. Currently, the compression-bending capacity of the circular hollow reinforcement concrete piers [12-13] and concrete wind turbine towers was predicted by several design codes [14-16]. In this design method of these codes, the assumption was adopted that the cross-section uncracking to make sure the relationship between compressive zone and tensile zone in the horizontal joint. Meanwhile, the compression-bending capacity is obtained by the principle of cross-sectional force balance [17]. It is clearly inappropriate to use these design methods to predict the compression-bending capacity of hybrid towers, because the changes in the cross-sectional flexural characteristics caused by the cracking of horizontal joints in concrete wind turbine towers have not been considered [18].

Several studies [19-22] developing design methods of stiffened steel tubes under the combined loading conditions of compression, bending, shear and torsion. However, these findings cannot directly apply to the prestressed concrete structures. Tan et al. [18] carried out four tests of compression-bending capacity of horizontal joints in prestressed concrete towers for wind turbines and analysed the results of the specimens in detail in terms of failure modes, compression-bending capacity, structural stiffness, bearing capacity degradation and prestressing force increment. However, due to limited experimental data, there is a lack of systematic exploration of the key parameters that influence compression-bending capacity. To summarise, the existing design methods were developed based on the bonded prestressed concrete structure, apart from to the actual configuration of concrete wind turbine towers (i.e. the equation of strain compatibility will not be satisfied between concrete and prestressed tendons, because the unbonded prestressed concrete structure was adopted), which will overestimate the compression-bending capacity of concrete wind turbine towers.

Therefore, this study aims to propose a more accurate design method to predict the compression-bending capacity of concrete wind turbine towers, with the consideration of the prestressing force increment. In this paper, a finite element (FE) model will firstly be developed and validated against four experimental results reported by Tan et al. [18]. Subsequently, a parametric study is conducted to generate a total of 125 numerical results, investigating key parameters in terms of concrete and steel reinforcement strength, cross-sectional diameters, wall thickness and

78 prestressing force. Finally, based on the newly generated numerical results, a new design method is
 79 proposed for the prestressed concrete towers subjected to compression and bending.

80 2 Summary of tests

81 A total of four specimens had been tested [18]. As illustrated in Fig 2, the specimen was comprised
 82 of a lower segment, an upper segment, and a transition segment. The horizontal joint was located
 83 between the upper and lower segment. Each specimen had a height of ($H = 3$ m), an outer diameter
 84 of ($D = 650$ mm), and a wall thickness of ($t = 75$ mm). The vertical prestressing force was provided
 85 by prestressed tendons anchored between the top of loading connection device and the bottom of
 86 foundation. The transverse load was applied at a distance ($H_1 = 2.6$ m) from the height of the
 87 horizontal joint. Meanwhile, the load was applied by the 100-ton hydraulic servo actuator. The
 88 average compressive strength of concrete was 80.1 MPa, and the average yield strength of the
 89 HRB400 steel reinforcement was 447 MPa.

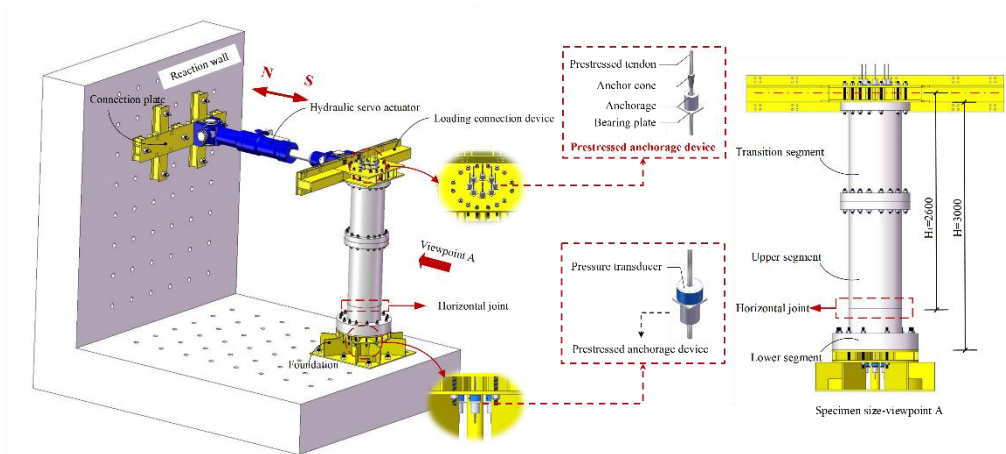


Fig 2 Schematic drawing of the loading device (unit: mm) [18].

90 The four specimens conducted primarily focus on two variables: the application of different
 91 prestressing loads and the application situation of epoxy resin at the horizontal joint. The specimen
 92 number and parameter design is presented in Table 1.

Table 1. Test parameters of specimens [18].

Specimen number	Precompression ratio	Application of epoxy resin on horizontal joint
PCT-1	0.3	No
PCT-2	0.15	No
PCT-3	0.3	Yes
PCT-4	0.15	Yes

3 Finite element modelling

The nonlinear finite element (FE) analysis package ABAQUS 6.14 [23] was utilized in the numerical investigation. The geometric and material nonlinearities are considered in the FE model. During the modelling process, the loading device and the loading of the experimental programme were reasonably simplified to enhance the computational efficiency and accuracy of the model.

3.1 Material properties

The constitutive curve proposed by Mander [24] was adopted for the elastic-plastic constitutive model of concrete, considering only the confinement effect of the outer layer of stirrups on the concrete material. The fracture energy of the concrete was calculated by *CEB-FIP MC90* [25]. Meanwhile, the tensile strength of the concrete was determined according to *GB 50010* [14]. An ideal elastic constitutive model was employed for the foundation and bolts, while an ideal elastic-plastic constitutive model was used for the steel reinforcement and prestressed tendons, as specified in *BS EN 1992-1-1* [26]. Furthermore, the expansion coefficients of 10^{-5} are applied to the prestressed tendons.

3.2 Contact properties

The “Embedded Region” was employed to simulate the interaction between the steel reinforcement and the concrete. Surface-to-surface contact was established at four interfaces: between the transition and upper segments, between the lower segment and the foundation, between the foundation and the ground, and at the horizontal joint interface. The tangential behaviour was defined using a penalty friction formulation with a friction coefficient of 0.5, while the normal behaviour was characterized as “hard contact”. Meanwhile, the horizontal joints of specimens PCT-3 and PCT-4 are connected using epoxy resin, and the adhesive behaviour of the epoxy resin is simulated using the cohesive behaviour. A reference point, RP-1 (see Fig. 3), was established at the mid-height of the loading connection device and coupled with the bolt holes located on the upper part of the transition segment. Another reference point, RP-2 (see Fig. 3), was established at the upper surface of the loading connection device for the prestressed tendon anchor and coupled with the upper position of the prestressed tendons and the upper surface of the transition segment. The loading connection device was simplified to a truss element and coupled with the transition segment. The bottom position of the prestressed tendons was coupled with the reserved holes in the

foundation.

3.3 Meshing and boundary conditions

The solid element (C3D8R) was utilized for the concrete structures. Truss element (T3D2) was used for the simulation for the steel reinforcement, bolts, the connection device and prestressed tendon. Shell elements (S4R) was employed for the foundation [27-29]. Following a mesh sensitivity analysis that balanced mesh density and computational time, a final mesh size of the transition segment, ground and the foundation was selected as 120×120 mm, while a mesh size of as 30×30 mm was chosen for the concrete structure and steel reinforcement in the upper segment and lower segment. The ground part and the loading connection device were considered as rigid body. Additionally, three translational degrees of freedom U1, U2 and U3 and three rotational degrees of freedom UR1, UR2 and UR3 were fixed for the bolt hole areas of the foundation part and the ground part. Transverse load was applied at reference point RP-1. The specific mesh and constraint settings for each component of the model are illustrated in Fig 3.

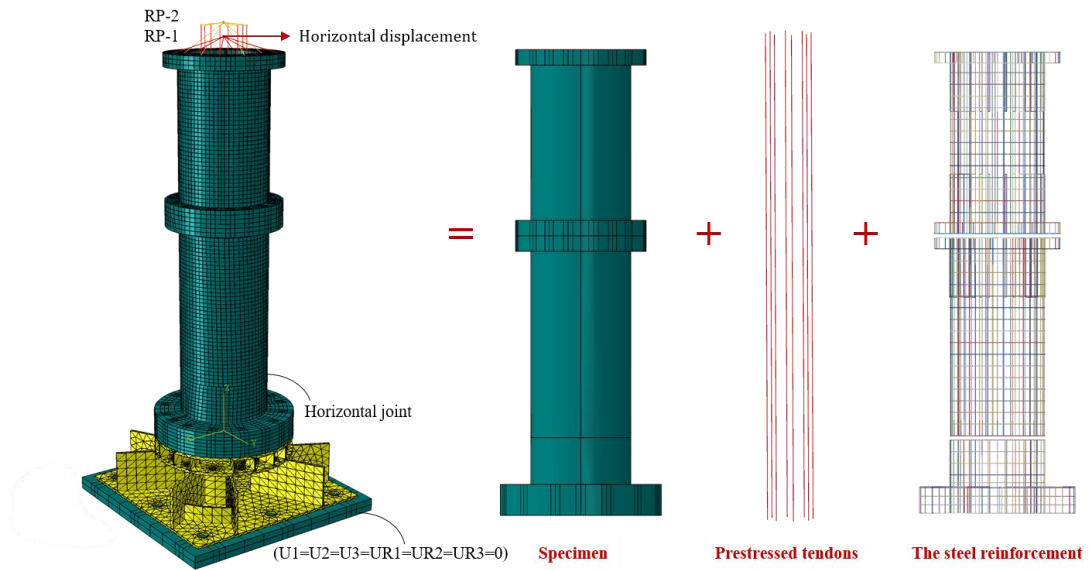


Fig 3 Detail of FE model.

3.4 Model validation

The load-displacement curves obtained from the FE model was compared against those experimental ones. As shown in Fig 4, the two sets of curves are in good agreement. Table 2 summarises the compression-bending capacities obtained by test and FE modelling. Results indicate that the FE simulations are close to the experimental values, with differences less than 5%. Fig 5 illustrates the failure modes of specimens PCT-1 and PCT-2. FE models successfully reproduced

the concrete crushing and cracking of horizontal joints observed in the test, indicating that the FE model developed in this study can effectively simulate the behaviour of concrete wind turbine towers.

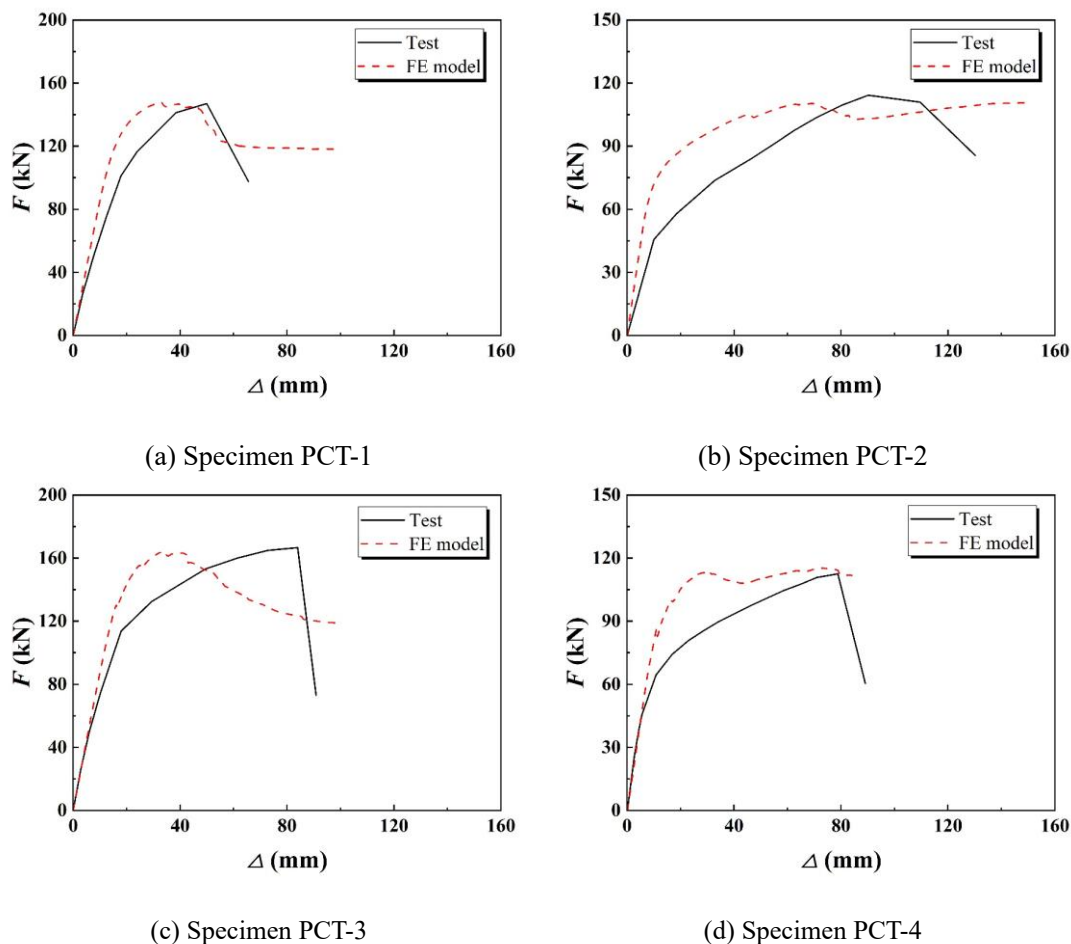
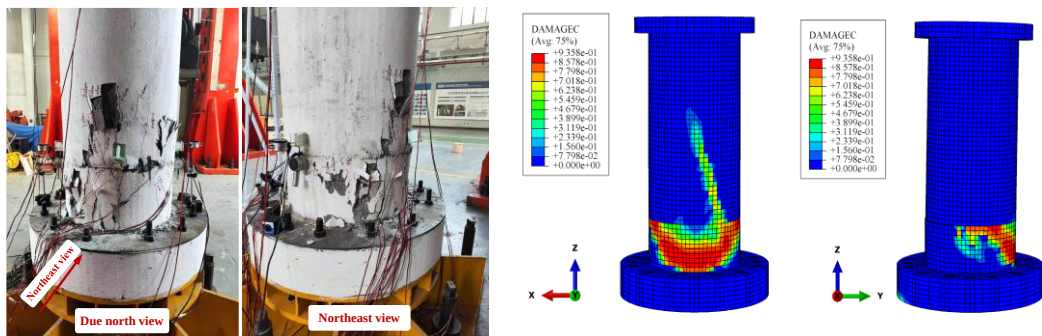


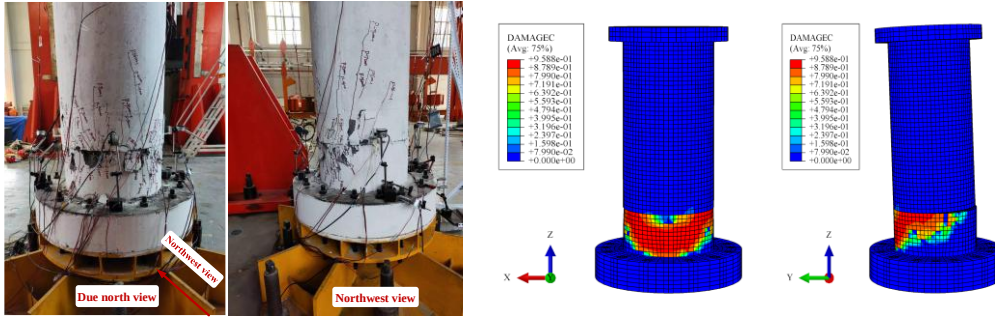
Fig 4 Comparison of load-displacement curves between FE model and test [18].

Table 2 Comparison of compression-bending capacities between FE model and test [18].

Specimen number	PCT-1	PCT-2	PCT-3	PCT-4
Test	147	118	167.3	112.4
FE	147.6	112.6	164	115.3
Test/ FE	0.996	1.048	1.02	0.975



(a) Specimen PCT-1



(b) Specimen PCT-2

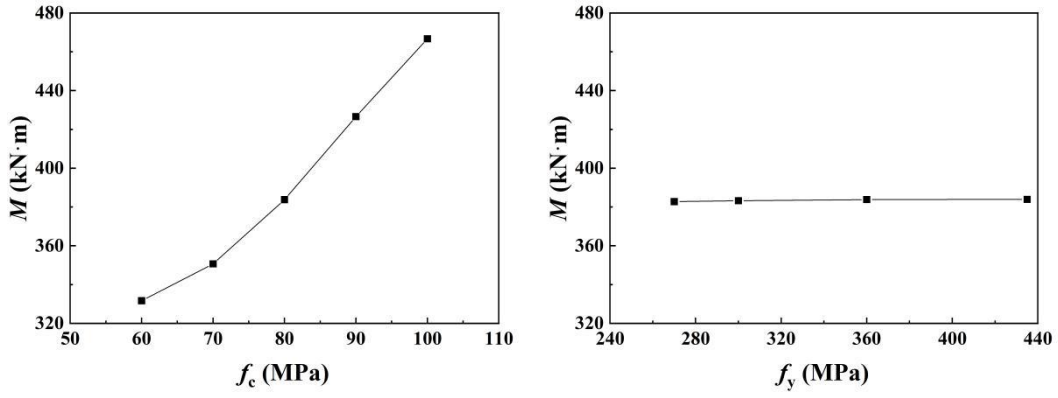
Fig 5 comparison of failure modes between FE model and test [18].

4 Parametric study

To investigate the factors influencing the compression-bending capacity of concrete wind turbine towers, the validated FE model was employed in the parametric study to extensively investigate key parameters including the strengths of concrete and steel materials, cross-section dimensions and prestressing force of the specimen. Based on the practical design of prestressed concrete towers, the minimum structural reinforcement ratio was adopted in FE model.

4.1 Effects of material strengths

The compressive strength of concrete (f_c) was considered from C50 to C90, and the yielding strength of the steel reinforcement (f_y) was considered as 235, 300, 360 and 435 MPa. The FE model with a diameter (D) of 650 mm and a wall thickness (t) of 75 mm was employed in parametric analysis. The FE results are shown in Fig 6. According to the analysis results, the yield strength (f_y) of the steel reinforcement has no effect on the compression-bending capacity. On the contrary, with the increase of the compressive strength of concrete, the compression-bending capacity increases consistently. That was probably because the compressive stresses are borne by the concrete material at the horizontal joint primarily. Meanwhile, the minimum structural reinforcement ratio was adopted at the cross-section of horizontal joints, the contribution of reinforcement to the compression-bending capacity is limited. Therefore, the yield strength of steel reinforcement has little influence on the compression-bending capacity of concrete wind turbine towers.



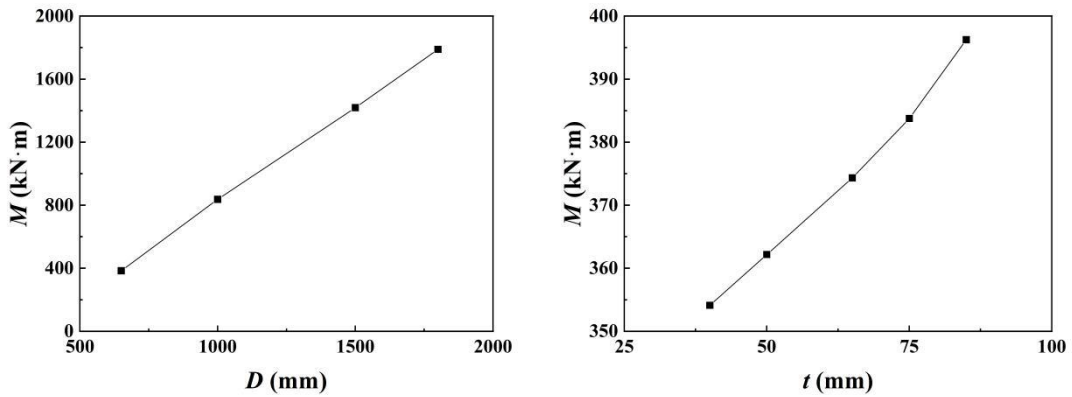
(a) The f_c - M curves

(b) The f_y - M curves

Fig 6 Effects of the material strength for compression-bending capacity.

4.2 Effects of cross-sectional dimensions

In the parametric study, the diameters (D) (650-1800 mm) and the wall thickness (t) (40-85 mm) were taken the same as experimental programme reported in Section 2. As shown in the Fig 7, with the cross-sectional diameters and wall thickness increasing, the compression-bending capacity keeps improving, which indicates the significance of cross-sectional dimensions on the resistance. The variation in wall thickness is expected to impact the restraint effect of stirrup for the concrete, because the double-layer stirrup was adopted for concrete wind turbine towers. Simultaneously, When the diameter or wall thickness of the cross-section increases, the second moment of inertia of the cross-section also increases, thereby enhancing the compression-bending capacity of specimens. Therefore, according to the analysis results, a diameter-thickness ratio ($\frac{D}{t}$), defined as the ratio of the cross-sectional diameters and wall thickness, is introduced herein to quantify the influence of the cross-sectional dimensions for the compression-bending capacity.



(a) The D - M curves

(b) The t - M curves

Fig 7 Effects of the geometric dimensions for compression-bending capacity.

179

180 4.3 Effects of prestressing force

181 Based on the prestressing force in the actual hybrid tower structure, the prestressing force is
 182 controlled using a precompression ratio [18], which is defined by Eq. (1). Fig 8 illustrates that the
 183 compression-bending capacity increases consistently, as the increase of the precompression ratio,
 184 within the precompression ratio range of 0.1 to 0.5. This may be due to the initial applied
 185 prestressing force affecting the relationship between the prestressing force at failure and the size of
 186 the compression zone of the horizontal joint. Therefore, the prestressing force is one of the key
 187 factors affecting the compression-bending capacity. In the current wind power industry, the
 188 precompression ratio range is typically set less than 0.4 to meet the fatigue requirements of hybrid
 189 tower structures. Therefore, the range is applied to subsequent analyses.

$$190 \quad \alpha_{py} = \frac{F_{py}}{f_{ck} \cdot A} \quad (1)$$

191 where α_{py} represents the precompression ratio of the specimens; F_{py} denotes the prestressing
 192 force sum of the specimen; f_{ck} indicates the characteristic strength of concrete cubes; A represents
 193 the full cross-sectional area of the horizontal joint in the specimen.

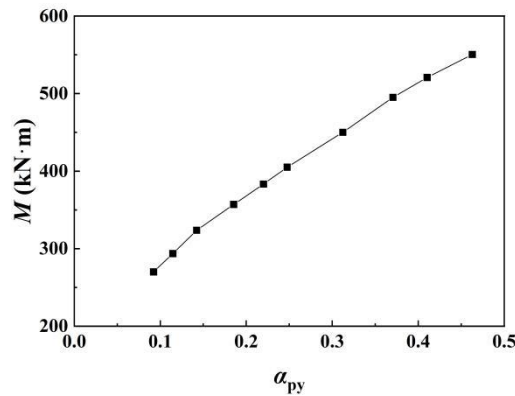


Fig 8 Effects of the precompression ratio for compression-bending capacity.

194 4.4 Effects of loading points of prestressing force

195 Fig 9 illustrates the relationship between the total prestressing force increment $\Delta F_{py,mean}$ of the
 196 prestressing force at ultimate and the prestressing force increment ΔF_{py} at the cross-sectional
 197 centreline of the horizontal joint. It was found that the vertical increment of the prestressing force
 198 at the centreline of the cross-section is equal to the mean total increment of the prestressing force in
 199 the horizontal joint cross-section, which is consistent with the experimental results [18]. It indicates

200 that the loading points of prestressing force still act at the centre of the horizontal joint cross-section.

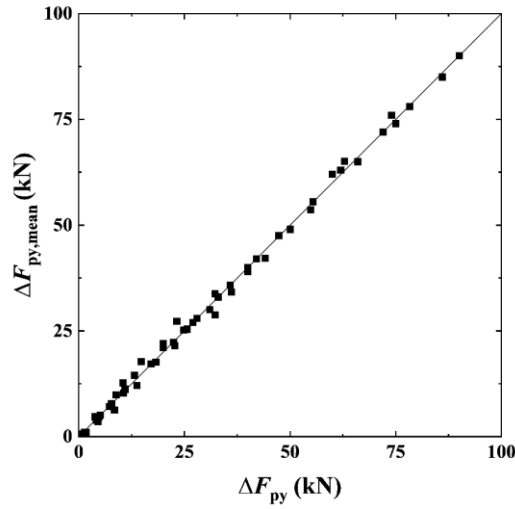


Fig 9 Loading points of prestressing force.

201 In summary, the key factors influencing the compression-bending capacity in prestressed
 202 concrete towers in terms of concrete strength, diameter-thickness ratio, and precompression ratio.
 203 When the cross-sectional geometric dimension and material properties are determined, the
 204 magnitude of the prestressing force is the only factor that affects the compression-bending capacity
 205 of the concrete wind turbine towers.

206 5 Existing design methods

207 5.1 GB/T 19072 [30]

208 GB/T 19072 [30] is a Chinese design standard for wind turbines-tower, with the design criteria
 209 specified in Chapter 7. In this design approach, the parameters related to steel reinforcement are
 210 omitted, because the minimum structural reinforcement ratio is adopted in prestressed concrete
 211 tower for wind turbines and the steel reinforcement in the horizontal joint is disconnected. the
 212 compressive strength parameter of the prestressed tendons is taken as 0, because the unbonded
 213 prestressed concrete structure is adopted in prestressed concrete wind turbine tower structure. The
 214 calculation of the compression-bending capacity (M) is calculated according to Eq. (2).

$$215 \quad M = \alpha_1 f_c A (r_1 + r_2) \frac{\sin \pi \alpha}{2\pi} + f'_{py} A_p r_p \frac{\sin \pi \alpha}{\pi} + (f_{py} - \sigma_{p0}) A_p r_p \frac{\sin \pi \alpha_t}{\pi} \quad (2)$$

216 where f_c is the concrete compressive strength; f'_{py} is the compressive strength of prestressed
 217 tendons; f_{py} is the tensile strength of prestressed tendons; r is the radius of cross-section; r_p is a
 218 representative for distribution radius of prestressed tendons; A is the cross-sectional area; A_p is a

delegate for the total cross-sectional area of prestressed tendon; σ_{p0} is the tensile stress of prestressed tendons after stretching. α_t is the half-angle coefficient of the tension zone; α is the half-angle coefficient of the compression zone.

The half-angle coefficient of tension zone (α_t) is calculated based on the half-angle coefficient of the compression zone (α), as given by Eq. (3), and when $\alpha \geq 2/3$, α_t is taken as 0.

$$\alpha_t = 1 - 1.5\alpha \quad (3)$$

Determine the half-angle coefficient of the compression zone (α) according to Eq. (4).

$$N = \alpha\alpha_1f_cA - \sigma_{p0}A_p + \alpha f'_{py}A_p - (1 - 1.5\alpha)(f_{py} - \sigma_{p0})A_p \quad (4)$$

where N is the gravity load of specimens.

5.2 T-CEC 5008 [31]

T/CEC 5008 [31] is a Chinese design standard. The design of wind turbine tower is presented in Chapter 6. In this design approach, the parameters are identical with *GB/T 19072* [30]. In addition, the bending contribution of the steel reinforcement is considered; The calculation of the compression-bending capacity (M) is calculated by Eq. (5).

$$M = (\alpha_1f_cAr + rf_yA_s + r_pf'_{py}A_p)\frac{\sin \alpha\pi}{\pi} + [(f_{py} - \sigma_{p0})A_p r_p + rf_yA_s]\frac{\sin \alpha_t\pi}{\pi} \quad (5)$$

where f_y is the yielding strength of the steel reinforcement; A_s is the total cross-section area of longitudinal reinforcement. The ratio between the half-angle coefficient in the compression zone and that in the tension zone is given in Eq. (6).

$$\alpha_t = 1 - 1.5\alpha \quad (6)$$

The half-angle coefficient of the compression zone (α) is calculated according to Eq. (7).

$$N = \alpha\alpha_1f_cA - \sigma_{p0}A_p + \alpha f'_{py}A_p - (1 - 1.5\alpha)(f_{py} - \sigma_{p0})A_p + (2.5\alpha - 1)f_yA_s \quad (7)$$

5.3 Design method proposed by Chen et al. [32]

Chen et al. [32] proposed a simplified method to estimate the ultimate strength of reinforced pipe segments. The calculation of the compression-bending capacity (M) is calculated by Eq. (8).

$$M = \sigma_m W \quad (8)$$

where σ_m is the stress caused by the moment, which is calculated according to Eq. (9).

$$\sigma_m = f_{cu} - \frac{N}{A} \quad (9)$$

where f_{cu} is the compressive strength of the steel fibre-reinforced concrete; A is the cross-sectional area; N is the gravity load of specimens.

5.4 Design method proposed by Tan et al. [18]

Tan et al. [18] proposed a compression-bending capacity calculation method in prestressed concrete towers for wind turbines. In this design approach, the parameters are identical with *GB/T 19072* [30], and the bending contribution of steel reinforcement within the tensile zone of horizontal joints is disregarded. The calculation of the compression-bending capacity (M) is calculated by Eq. (10).

$$M = (\alpha_1 f_c A + f_y A_s) r \cdot \frac{\sin \pi \alpha}{\pi} + \sigma_{py} \cdot A_p \cdot r_{py} \cdot \left(\frac{\sin \pi \alpha_t}{\pi} - \frac{\sin \pi \alpha}{\pi} \right) \quad (10)$$

The half-angle coefficient in the compression zone (α) is given in Eq. (11).

$$\alpha = \alpha_{py} \quad (11)$$

The half-angle coefficient of tension zone (α_t) is calculated based on the half-angle coefficient in the compression zone (α) affecting the horizontal joint, as given by Eq. (12).

$$\alpha + \alpha_t = 1 \quad (12)$$

6 Proposed design method

6.1 Theoretical analysis

The geometric characteristics of the horizontal joint sections is shown in Fig 10.

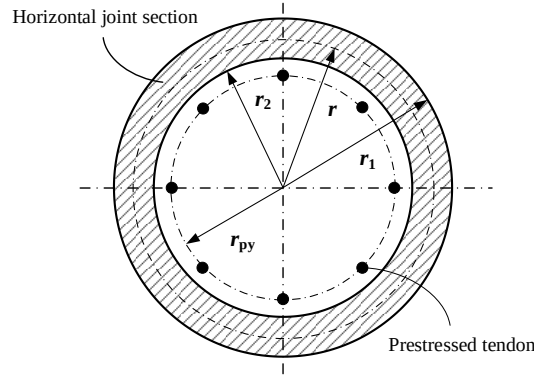


Fig 10 Horizontal joint cross-sectional geometric properties.

The cross-sectional area A and cross-sectional radius r of horizontal joints are calculated by Eq. (13) and Eq. (14), respectively.

$$A = \pi(r_1^2 - r_2^2) \quad (13)$$

$$r = \frac{r_1 + r_2}{2} \quad (14)$$

where r_1 denotes the external radius of the tower tube cross-section; r_2 indicates the internal radius of the tower tube cross-section.

6.1.1 Elastic stage

Before the horizontal joints cracking, the cross-sectional stress distribution is analysed with reference to the flexural of thin-walled circular hollow section beams. The stress state of the horizontal joint at the elastic stage is shown in Fig 11.

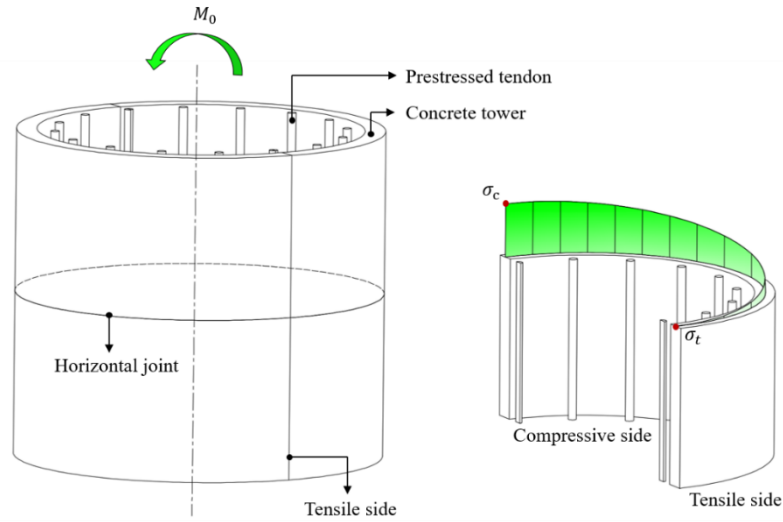


Fig 11 Schematic diagram of stress model during critical separation phase of the horizontal joint.

The stresses on the tension side (σ_t) and compression side of the horizontal joint (σ_c) are calculated according to Eq. (15) and Eq. (16), respectively. At the critical cracking stage of the horizontal joints, the stress on the tension side edge of the horizontal joint is exactly zero, which is denoted as $\sigma_t = 0$.

$$\sigma_t = \frac{F_z + F_{py}}{A} - \frac{M_0}{W_z} \quad (15)$$

$$\sigma_c = \frac{F_z + F_{py}}{A} + \frac{M_0}{W_z} \quad (16)$$

where W_z is the second moment of inertia of circular section calculated by Eq. (17); M_0 is the critical bending moment of the horizontal joint cracking given by Eq. (18).

$$W_z = \frac{\pi r_1^3}{4} \left(1 - \left(\frac{r_2}{r_1} \right)^4 \right) \quad (17)$$

$$M_0 = \frac{F_z + F_{py}}{A} \cdot \frac{\pi r_1^3}{4} \left(1 - \left(\frac{r_2}{r_1} \right)^4 \right) \quad (18)$$

6.1.2 Plastic failure stage

As the bending moment applied to horizontal joints increases, cracks will occur at the horizontal joint. Therefore, the compression zone of the horizontal joint decreases, eventually leading to crushing of concrete and yielding of steel reinforcement on the compression side of the horizontal joint. The stress pattern at ultimate is shown in Fig 12.

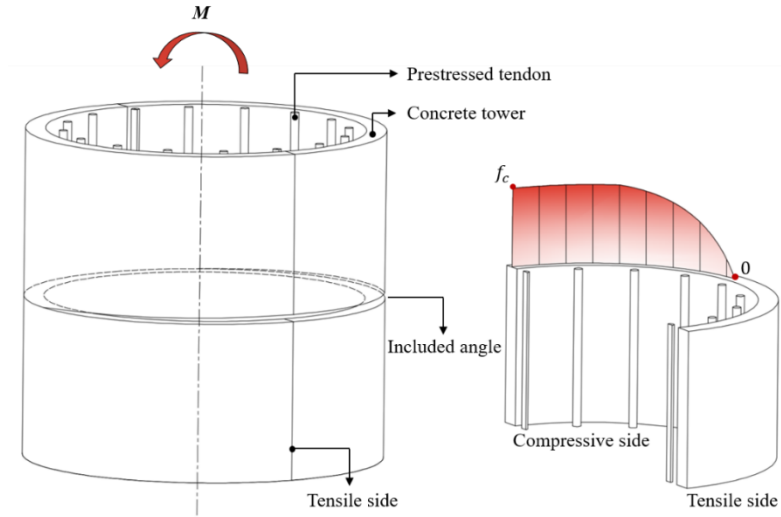


Fig 12 Schematic diagram of the failure under compression and bending condition.

Based on the experimental results and the parametric study, it is observed that when the horizontal joint reaches its ultimate compression-bearing capacity, the loading point of the prestressing force remains largely unchanged, and the contact area of the horizontal joint cross-section significantly decreases, which leads to a shift in the neutral axis of the cross-section. Additionally, there is a substantial increment of the prestressing force [18]. This assumption is applied that the stress distributions of between concrete and prestressed tendons are simplified to rectangular distributions, and the stress pattern of the horizontal joint cross-section is shown in Fig 13.

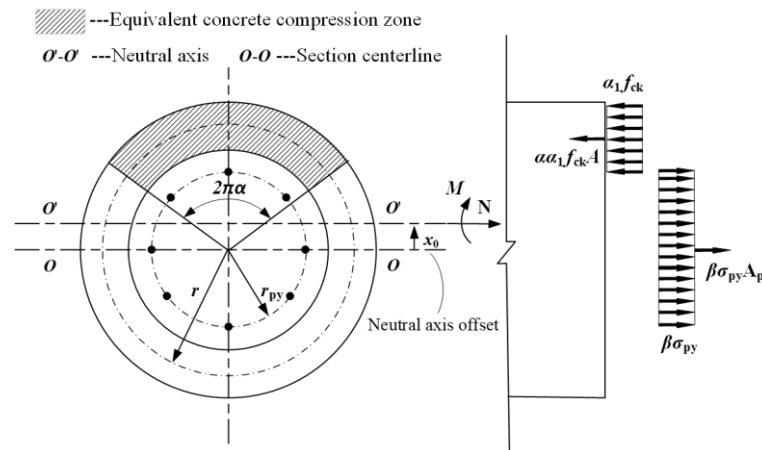


Fig 13 Calculation diagram for the compression-bending capacity.

The calculation of the compression-bending capacity (M) is calculated according to Eq. (19):

$$M = \alpha_1 f_{ck} A r \frac{\sin \pi \alpha}{\pi} \quad (19)$$

where α_1 is taken as 1.0 when the concrete strength grade does not exceed C50, and as 0.94

for a concrete strength grade of C80, with linear interpolation used in between; α represents the half-angle coefficient of compression zone is calculated by Eq. (20);

$$\alpha = \frac{F_z + \beta \sigma_{py} A_p}{\alpha_1 f_{ck} A} \quad (20)$$

where F_z is the gravity load of specimens; A_p is the total area of prestressing tendons; σ_{py} is the internal stress of prestressed tendons; β represents the increment coefficient of prestressing force.

6.2 Determination of coefficients in design formula

When the horizontal joint cracks, the prestressed tendons were stretched, resulting in an increment in prestressing force. The increment of prestressing force was determined by the included angle of the horizontal joint after cracking. When the moment bending reached the compression-bending capacity of horizontal joints, the strain of the concrete on the compression side has already reached the ultimate strain. The ultimate strain of the concrete was defined by concrete strength and was also influenced by the stirrup-confined. Furthermore, the confinement effect of the stirrups on the concrete was affected by the wall thickness of horizontal joints. The strain of concrete was influenced by the diameter of the horizontal joint, because changes in the cross-sectional diameter and wall thickness directly influence the second moment of inertia of the cross-section. Therefore, according to the analysis results, a characteristic coefficient (φ), defined as the product of the precompression ratio, diameter-thickness ratio, and the ratio of concrete strength, is introduced herein to quantify the influence of the precompression ratio, cross-sectional dimensions, and concrete strength for the prestressing force increment. The characteristic coefficient (φ) is calculated by Eq. (21)

$$\varphi = \alpha_{py} \left(\frac{D}{t} \right) \left(\frac{f_{ck}}{f_{ck,60}} \right), (10 \leq \frac{D}{t} \leq 50; 39 \leq f_{ck} \leq 58; 0.1 \leq \alpha_{py} \leq 0.4) \quad (21)$$

Fig 14 illustrates the functional relationship between the increment coefficient of prestressing force (β) and the characteristic coefficient (φ). The increment coefficient of prestressing force (β) is calculated by Eq. (22) based on regression curve fitting.

$$\beta = \frac{2.2}{(\varphi+1)^2} + 1, (1 \leq \varphi \leq 30) \quad (22)$$

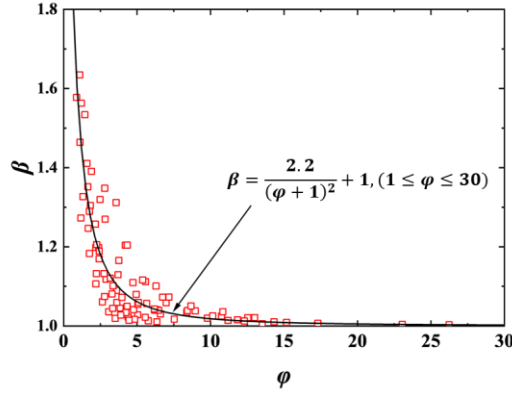


Fig 14 Increment coefficient of prestressing force.

6.3 Calculation procedure

The calculation process to determine the compression-bending capacity of concrete wind turbine towers has three steps.

Step 1: Determine the initial precompression ratio (α_{py}) and the characteristic coefficient (φ) according to Eqs. (23) and (24).

$$\alpha_{py} = \frac{F_{py}}{f_{ck} \cdot A} \quad (23)$$

$$\varphi = \alpha_{py} \left(\frac{D}{t} \right) \left(\frac{f_{ck}}{f_{ck,60}} \right), (10 \leq \frac{D}{t} \leq 50 ; 39 \leq f_{ck} \leq 58 ; 0.1 \leq \alpha_{py} \leq 0.4) \quad (24)$$

Step 2: Determine the increment coefficient of prestressing force (β) according to Eq. (25).

$$\beta = \frac{2.2}{(\varphi+1)^2} + 1, (1 \leq \varphi \leq 30) \quad (25)$$

Step 3: The half-angle coefficient of compression zone (α) is calculated based on the increment coefficient of prestressing force (β) affecting the horizontal joint, as given by Eq. (26).

$$\alpha = \frac{F_z + \beta \sigma_{py} A_p}{\alpha_1 f_{ck} A} \quad (26)$$

Step 4: Calculate the compression-bending capacity (M) of concrete wind turbine towers, as given by Eq. (27).

$$M = \alpha_1 f_c A r \frac{\sin \pi \alpha}{\pi} \quad (27)$$

7. Result comparison

Fig 15 illustrates the relationship between the compression-bending capacity of the horizontal joint obtained from the FE simulation and the calculation method proposed in this paper. The predicted results of compression-bending capacity are in good agreement with numerical (and experimental) results with good accuracy and consistency and the error less than 10%.

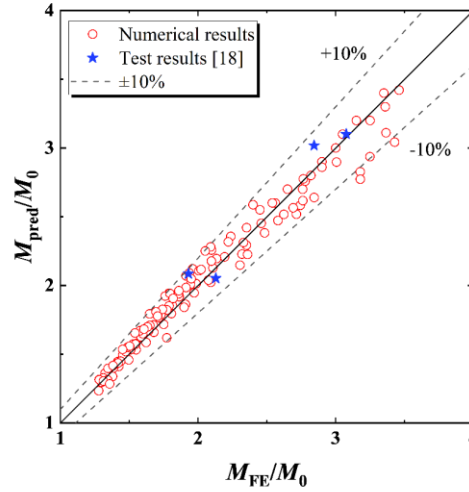


Fig 15 Comparison between predicted results and numerical (and experimental) results.

A comparison of the predicted compression-bending capacities using existing and proposed design methods was conducted against the experimental and numerical results. Fig 16 illustrates the relationship between the compression-bending capacity from the FE parametric analysis and those calculated using different codes herein.

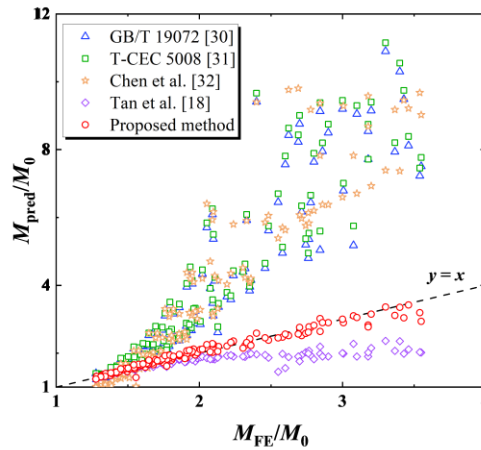


Fig 16 Comparison of predicted results with existing and proposed design method.

The comparisons clearly indicate that the predictions in terms of *GB/T 19072* [30], *T-CEC 5008* [31], and literature [32] are unsafe. However, the prediction by literature [18] was on the conservative side. It is clearly shown that the existing design method cannot accurately and reliably assess the compression-bending capacity of concrete wind turbine towers. Furthermore, Table 3 summarizes the comparison of design method based on all FE results. The new design method provides rather accurate and consistent predictions with compression-bending capacity a mean value of 1.01 and coefficient of variation (CoV) of 0.045, which can serve as a design reference in concrete wind turbine towers.

Table 3 Comparison of design method between predicted results and simulated results.

Design method	Mean	CoV
<i>GB/T 19072</i> [30]	1.73	0.762
<i>T-CEC 5008</i> [31]	1.78	0.771
Chen et al. [32]	1.76	0.866
Tan et al. [18]	0.84	0.371
This study	1.01	0.045

8 Conclusions

In this study, a finite element model was developed and validated against experimental results in terms of failure modes, load-displacement curves and compression-bending capacity. In the parametric study, a total of 125 numerical results were generated to consider three key parameters affecting the compression-bending capacity of prestressed concrete towers, including concrete strength, yield strength of steel reinforcement, cross-sectional diameters and the wall thickness, and prestressing force. Numerical results show that: (1) with the increase of the concrete compressive strengths, the compression-bending capacity in concrete wind turbine towers increase linearly; (2) the compression-bending capacity increase as the diameters and wall thickness of horizontal joints increase; (3) with the increase of the initial precompression ratio, the compression-bending capacity improves significantly. Based on the numerical results generated from parametric study, the accuracy and applicability of the existing design methods for compression-bending capacity were evaluated. These design methods were found to be either unsafe or overly conservative because the influence of the prestressing force increment and the change of the cross-sectional flexural characteristics was disregard. Therefore, a new design method is proposed, which could better reflect the impact of different initial prestressing force on the compression-bending capacity of concrete wind turbine towers. By comparing the predicted results with experimental and numerical results, the proposed design method is found to be accurate and consistent, with the mean value of the experimental-to-experimental (and numerical) results being 1.01 and the corresponding CoV being 0.045.

Author contributions

Ji-Ke Tan: Conceptualization, Methodology, Investigation and Review & Editing. Wan-Lang Zheng: Investigation and Writing-Original Draft. Yu-Hang Wang: Supervision. Mei-Ni Su: Supervision and

Review & Editing. Lang Zhao: Supervision. Yun-Hui Zhang: Investigation and Review & Editing.
Yong Feng: Supervision. De-Xiong Chen: Supervision.

Declaration of interest

No conflict of interest, financial or otherwise.

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References

- [1] Lin B, Li Z. Towards world's low carbon development: the role of clean energy. *Applied Energy*, 2022, 307:118160.
- [2] Hernandez-Estrada E, Lastres-Danguillecourt O, Robles-Ocampo J B, et al. Considerations for the structural analysis and design of wind turbine towers: A review. *Renewable and Sustainable Energy Reviews*, 2020, 137(8).
- [3] Lana J, PAAM Júnior, Magalhes C A, et al. Behavior study of prestressed concrete wind-turbine tower in circular cross-section. *Engineering Structures*, 2021, 227:111403.
- [4] Li Z Y, Xu B, Wang J, et al. Experimental and two-scale numerical studies on the behavior of prestressed concrete-steel hybrid wind turbine tower models. *Engineering Structures*, 2023, 279.
- [5] Cheng Y, Zhao Y N, Qi H T, et al. Intelligent optimal design of steel-concrete hybrid wind turbine tower based on evolutionary algorithm. *Journal of Constructional Steel Research*, 2024, 218:108729.
- [6] Gkantou M, Martinea-Vazquez P, Baniotopoulos C. On the structural response of a tall hybrid onshore wind turbine tower. *Procedia Engineering*, 2017, 199:3200-3205.
- [7] Chen J, Li J, He X. Design optimization of steel-concrete hybrid wind turbine tower based on improved genetic algorithm. *The Structural Design of Tall and Special Buildings*, 2020,

29(10):e1741.

- [8] Cheng Y, Cao L, Liu J, et al. Intelligent analysis of dynamic characteristics of steel-concrete hybrid wind turbine tower based on adaptive vibration mode. *Structures*, 2024, 68:107235.
- [9] Schaarschmidt D, Flederer H. Vorgespannte Trockenfugen für modulare Brücken unter zyklischer Belastung. *Beton-und Stahlbetonbau*, 2021, 116(5):378-386.
- [10] Jin K Y, Wang Y H, Zhou X H, et al. Behavior of super-sized thin-walled CFDST columns for wind turbine towers subjected to combined loads: Experiment. *Engineering Structures*, 2024, 303:117458.
- [11] Zhou Z, Zhou X H. Mechanical behavior of prestressed UHPC wind turbine tower columns under combined axial compression and bending. *Structural Concrete*, 2024, suco.202400223.
- [12] Wang Z, Wang J Q, Liu T X, et al. An explicit analytical model for seismic performance of an unbonded post-tensioned precast segmental rocking hollow pier. *Engineering Structures*, 2018, 161:176-191.
- [13] Li Z X, Du C Y, Liang, X, et al. Flexural behavior of circular hollow RC piers with reduced amounts of inner hoops. *International Journal of Concrete Structures and Materials*, 2020, 14(9):1-15.
- [14] Code for design of concrete structures: GB 50010-2010. Beijing: China Architecture & Building Press, 2015.
- [15] Code for Design of Concrete-Steel Hybrid Tower of Wind Turbine: NBT 10907-2021. Beijing: China Water & Power Press, 2022.
- [16] Standard for design of high-rising structures: GB 50135-2019. Beijing: China Planning Press, 2019.
- [17] Chen H P, Jiang Y, Li W B. Residual load-bearing capacity of corroded reinforced concrete columns with an annular cross-section. *Ocean Engineering*, 2023, 286 (P1):115519.
- [18] Tan J K, Zheng W L, Wang Y H, et al. Study on the compression-bending capacity of horizontal joints in prestressed concrete towers for wind turbines. *Structures*, 2025, 72:108292.
- [19] Ren W, Zhou X H, Wang S J, et al. Pull-out behaviour of the pre-installed lifting anchor with supplementary reinforcement for the precast concrete segments of wind turbine hybrid towers. *Engineering Structures*, 2024, 303:117547.
- [20] Ren W, Zhou X H, Deng R, et al. Experimental and analytical investigation of stiffened steel tubes for wind turbine towers under compression-bending load. *Journal of Constructional Steel Research*, 2022, 193:107297.
- [21] Ren W, Zhou X H, Lin F Z, et al. Static behaviour of large-diameter stiffened steel tubes for wind turbine towers under combined compression-bending-torsion load. *Thin-Walled Structures*, 2022, 175:109272.

- 448 [22] Zhou Z, Chen C, Shen X J, et al. Conceptual design of a prestressed precast UHPC-steel hybrid
449 tower to support a 15 MW offshore wind turbine. *Engineering Structures*, 2024, 321:118939.
- 450 [23] ABAQUS, ABAQUS/Standard Version 6.14 User's Manual vols. I-III, Dassault Systèmes
451 Simulia Corp, Rhode Island, USA, 2014.
- 452 [24] Mander, J B, Priestley, M J, & Park R. Theoretical stress-strain model for confined concrete.
453 *Journal of structural engineering*, 1988, 114 (8):1804-1826.
- 454 [25] CEB-FIP model code. High-strength Concrete State of the Art Report. London: Thomas Telford,
455 1990.
- 456 [26] Eurocode 2: Design of concrete structures Part 1-1: General rules and rules for buildings: BS
457 EN 1992-1-1 2014. 2014.
- 458 [27] Zeng W Y, Ayough P, Zhou X H, et al. Performance of circular CFST columns strengthened
459 with CFRP grid-reinforced ECC under axial compression: Numerical modelling and design.
460 *Engineering Structures*, 2025, 325:119375.
- 461 [28] Wang Y H, Zeng W Y, Zhou X H, et al. Performance of hollow ultra-high-performance
462 concrete-high strength steel columns for wind turbine towers with different confinement forms
463 under compression-bending load. *Thin-Walled Structures*, 2025, 208:112749.
- 464 [29] Ayough P, Wang Y H, Zeng W Y, et al. Numerical investigation and design of UHPC-encased
465 CFST stub columns under axial compression. *Engineering Structures*, 2024, 302:117387.
- 466 [30] Wind turbines-Tower: GB/T 19072-2022. Beijing: Standards Press of China, 2022.
- 467 [31] Code of prestressed precast concrete tower for wind turbine: T/CEC 5008-2018. Beijing: China
468 Electric Power Press, 2018.
- 469 [32] Chen C, Shen X J, Zhou Z, et al. Feasibility study of a steel-UHPFRC hybrid tower for offshore
470 wind turbines, *Ocean Engineering*, 2023, 288:116140.